Telemetry Protocol v1

Datasheet and user note | Edge Telemetry Gateway

Application protocol that runs inside a TCP stream between one gateway and its clients. All multi-byte values are **big-endian**. Fields are read byte by byte, never by casting the receive buffer to a struct.

1. Frame layout

Offset (bytes)	Size	Field	Notes
0	1	magic	`0x78`
1	1	version / reserved	low nibble = version (`0b0001`), high nibble = reserved, must be 0
2-3	2	seq	u16
4-5	2	len	u16, payload bytes, 0 to 1024
6-7	2	type	u16, see section 2
8 ...	N	payload	N = len
8+N ...	4	crc32	see below

- Header is 8 bytes and the CRC adds 4. A frame is **12 + N** bytes. The largest frame is **1,036** bytes.
- **CRC-32:** IEEE 802.3, reflected polynomial `0xEDB88320`, initial value `0xFFFFFFFF`, reflected input and output, final XOR `0xFFFFFFFF`. It covers bytes 0 to 7+N, so the header is included. Test vector: the CRC of ASCII `"123456789"` is `0xCBF43926`.
- **seq** counts TELEMETRY frames. It starts at 0 on each connection and wraps at 65,536. On the other types it carries a number that labels the message (section 2).

2. Message types

Type	Value	Direction	Header seq	len	Sent when
TELEMETRY	0x0058	gateway to client	frame number	24	every sample tick (default 100 Hz)
HEARTBEAT	0x00BC	both	0	0	a side has sent nothing for one heartbeat interval
RESEND	0x001A	client to gateway	the seq wanted	0	the client sees a gap or a failure
ACK	0x0064	gateway to client	the seq requested	0	the gateway has the frame; the frame follows ACK immediately
REJECTED	0x006E	gateway to client	the seq it cannot supply	0	the frame is no longer in the gateway's buffer
ERROR	0x0005	client to gateway	0	0	the client would need a third RESEND within one second

Numbering: the high nibble of the low byte gives the direction. 0-4 client to gateway, 5-9 gateway to client, A-F both. The type field is 16 bits, so the high byte is 0x00.

TELEMETRY payload (24 bytes)

Offset	Size	Field	Notes
0	8	timestamp	u64, nanoseconds, BBB monotonic clock, at sample time
8	2	dev_id	u16
10	2	dev_addr	u16
12	6	accel x, y, z	i16 x 3, raw counts
18	6	gyro x, y, z	i16 x 3, raw counts

Signed values go on the wire as two's-complement bit patterns. The client converts counts to g and degrees per second. A TELEMETRY frame is 8 + 24 + 4 = **36 bytes**. HEARTBEAT, RESEND, ACK, REJECTED and ERROR have empty payloads, so each is a 12-byte frame.

3. Receiver rules

Condition	Action
fewer than 8 bytes buffered	wait for more
magic, version or reserved wrong; len above 1024; CRC mismatch; unknown type*	failure: discard the buffer, send RESEND(expected seq), scan for ACK or REJECTED. No allocation for an oversize len.
TELEMETRY seq later than expected	send RESEND for the missing frames
a third RESEND needed within one second	send ERROR, close, reconnect
peer silent for 3 x heartbeat interval	treat as dead, close
gateway receives a bad frame from a client*	close

* Defaults assumed during design review. Not yet confirmed.

4. Recovery and liveness

- **Gap:** the client expects 90 and receives 100. It sends RESEND for the missing frames and loops until it is back in step with the gateway.
- **Gateway reply:** ACK(k), then TELEMETRY frame k. If the frame is gone, REJECTED(k). The client logs the gap and continues.
- **Realignment:** after a failure the client discards its buffer and sends RESEND(expected seq). ACK and REJECTED have empty payloads, so each is a complete 12-byte frame with its own CRC. The client scans for a candidate whose magic matches, whose type is ACK or REJECTED, whose seq matches the request, and whose CRC verifies. The frame after the ACK starts exactly where the ACK ends.
- **ACK timeout:** 300 ms. On a timeout the client sends RESEND again*, and that counts toward the limit below.
- **Limit:** up to 2 RESENDs are allowed within any one second, for any frames. Needing a third means the client sends ERROR, closes and reconnects. The gateway closes when it receives ERROR.
- **Heartbeat:** each side sends HEARTBEAT if it has been idle for the interval. The gateway streams at 100 Hz, so in practice the client's heartbeats keep the gateway informed. The interval is a constant shared by both sides, since nothing negotiates it. The earlier draft's default was 1,000 ms.
- **Gateway buffer:** a fixed ring buffer of the last 16 TELEMETRY payloads of 24 bytes each, 384 bytes. On a replay the gateway rebuilds the header from the slot's seq and recomputes the CRC.

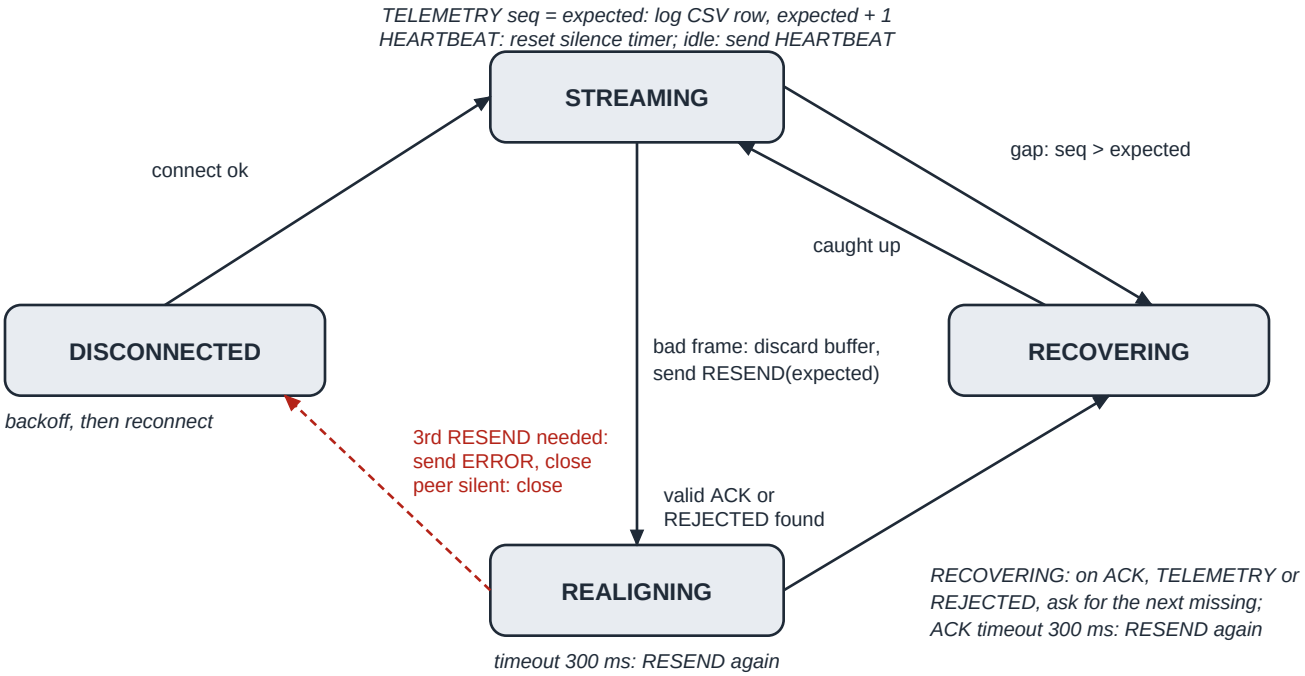
5. Constants

Name	Value
magic	0x78
version	0b0001 (low nibble of byte 1)
MAX_PAYLOAD	1024 bytes
Largest frame	1036 bytes
TELEMETRY frame	36 bytes
Control frame (empty payload)	12 bytes
Sample rate default	100 Hz
ACK timeout	300 ms
RESEND limit	2 per 1 second window, third triggers ERROR
Ring buffer	16 x 24 bytes = 384 bytes

6. State machines

Client states are per connection. The gateway keeps one entry per connected client. Both follow the rules in sections 2 to 4.

Client

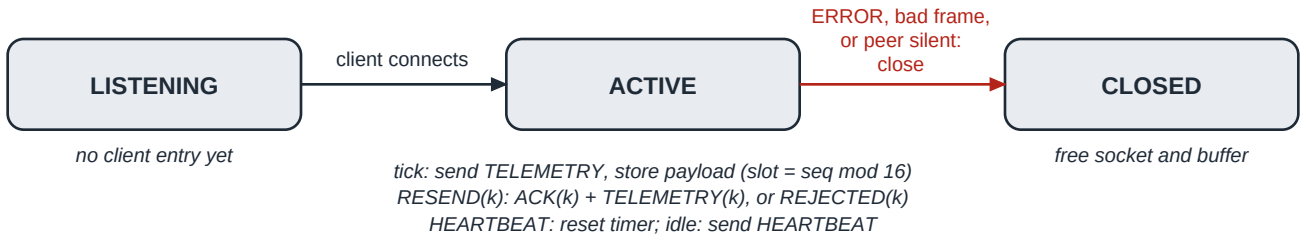


Dashed red arrows are available from every state. Every RESEND, including those sent after a timeout, counts toward the limit of 2 within one second.

State	Event	Action	Next
DISCONNECTED	backoff done	connect	STREAMING (expected seq = 0)
STREAMING	TELEMETRY, seq = expected	log CSV row, expected + 1	STREAMING
STREAMING	TELEMETRY, seq above expected	RESEND(expected)	RECOVERING
STREAMING	bad frame (magic, version, reserved, len above 1024, CRC, unknown type)	discard buffer, RESEND(expected)	REALIGNING
STREAMING	HEARTBEAT	reset silence timer	STREAMING
STREAMING	idle for one heartbeat interval	send HEARTBEAT	STREAMING
RECOVERING	ACK(k)	wait for TELEMETRY(k)	RECOVERING
RECOVERING	TELEMETRY(k)	log, expected + 1; if more missing, RESEND(next)	RECOVERING
RECOVERING	REJECTED(k)	log the gap, expected + 1; if more missing, RESEND(next)	RECOVERING
RECOVERING	caught up to the frame that triggered recovery	none	STREAMING
RECOVERING	ACK timeout (300 ms)	RESEND again	RECOVERING
REALIGNING	CRC-valid ACK or REJECTED for the requested seq	continue as in RECOVERING	RECOVERING

State	Event	Action	Next
REALIGNING	scan timeout (300 ms)	RESEND again	REALIGNING
any	a RESEND would be the 3rd within 1 s	send ERROR, close	DISCONNECTED
any	peer silent for 3 x heartbeat interval	close	DISCONNECTED

Gateway (one entry per connected client)



State	Event	Action	Next
(listening)	client connects	create the entry: socket, buffer, seq = 0, empty ring buffer	ACTIVE
ACTIVE	sample tick	send TELEMETRY(seq), store the payload in slot seq mod 16, seq + 1	ACTIVE
ACTIVE	RESEND(k), k in the ring buffer	send ACK(k), then TELEMETRY(k) rebuilt from the slot	ACTIVE
ACTIVE	RESEND(k), k not in the buffer	send REJECTED(k)	ACTIVE
ACTIVE	HEARTBEAT	reset silence timer	ACTIVE
ACTIVE	idle for one heartbeat interval	send HEARTBEAT	ACTIVE
ACTIVE	ERROR received	close	CLOSED
ACTIVE	bad frame from the client*	close	CLOSED
ACTIVE	peer silent for 3 x heartbeat interval	close	CLOSED
CLOSED	entry freed	free the socket and the buffer	gone

A replay is handled as an event inside ACTIVE, so the gateway holds no extra state per request. * Default assumed during design review.